TCS 424 — Satellite Communications

Exam-Focused Study Guide (covers every question in your past papers)

Compiled for Balqees

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How to use this guide

Your past papers cover two kinds of material:

1. **Things your uploaded slides already teach** — Multiple Access (FDMA/TDMA/CDMA), multiplexing (FDM/TDM/WDM). These sections are expanded and exam-formatted below.
2. **Things NOT in any file you sent me** — orbital mechanics, look angle, earth station architecture, transponders, TT&C, link budgets, routing, satellite history, configurations, demand assignment. I have written these from scratch, in the style your questions are asked, including **fully worked numerical solutions using your exact past-paper numbers** so you can see the method end-to-end.

Read in order. Sections 8–11 (link budget and look angle) carry the heaviest marks and are the most “mechanical” — master the formulas and you can solve any variant.

1. How Satellite Communication Works (Big Picture)

A communication satellite is essentially a **repeater in the sky**. The whole system has two segments:

- **Space segment**: the satellite itself — its bus (structure, power, propulsion, TT&C) and its payload (the transponders that actually do the communication job).
- **Ground segment**: earth stations (transmit/receive terminals) and the network/satellite control center that manages the satellite's health and position.

Basic sequence of events:

1. An earth station **uplinks** a signal (voice, data, video) modulated onto a carrier at the uplink frequency.
2. The satellite's **transponder** receives this weak signal, filters it, amplifies it, and **frequency-translates** it down to a different (downlink) frequency — this frequency shift is essential so the powerful outgoing signal doesn't swamp the sensitive incoming receiver.
3. The satellite **retransmits (downlinks)** the amplified signal back to Earth, where it is received by one or many earth stations spread over the satellite's coverage (footprint).

Because the satellite is so far away (up to ~36,000 km for GEO), signals lose enormous amounts of power (free-space path loss) and the round-trip introduces propagation delay (~240 ms for GEO one-way). Nearly every calculation-style question in this course exists to quantify these losses and delays.

Why satellites are useful: wide-area coverage without laying cable, ability to broadcast to many receivers simultaneously, and rapid deployment to remote/rural areas.

1.1 Brief History of the Satellite Industry

Year	Milestone
1957	**Sputnik 1** (USSR) — first artificial satellite; proved orbital flight was possible
1960	**Echo 1** (NASA) — first **passive** communications satellite; a metallized balloon that simply reflected radio signals
1962	**Telstar 1** — first **active** communications satellite; relayed the first live transatlantic TV signal
1963	**Syncom 2/3** — first satellites in **geostationary** orbit

Year	Milestone
1965	**Intelsat I (“Early Bird”)** — first commercial GEO communications satellite
1970s–80s	Growth of INTELSAT, INMARSAT (maritime), domestic satellite systems; rise of **VSAT** networks
1990s	Direct-to-home (DTH) TV broadcasting (DVB-S), mobile satellite systems
2000s–present	Broadband Ka-band systems, big **LEO constellations** (Iridium, and more recently Starlink/OneWeb) for global broadband and low-latency links

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2. Satellite Orbits: LEO, MEO, GEO

Satellites are classified by **altitude**, which determines coverage area, delay, and the number of satellites needed for continuous global coverage.

Feature	LEO (Low Earth Orbit)	MEO (Medium Earth Orbit)	GEO (Geostationary Orbit)
Altitude	~500–2,000 km	~2,000–35,786 km (typically ~10,000–20,000 km)	35,786 km (fixed)
Orbital period	~90–120 minutes	~2–12 hours	24 hours (matches Earth’s rotation)
Appears from ground	Moves rapidly across sky	Moves, slower than LEO	**Stationary** — always in the same spot
Propagation delay (one-way)	Very low (~1–4 ms)	Moderate (~25–70 ms)	High (~240 ms)
Coverage per satellite	Small — needs a large constellation for global coverage	Medium	Very large (~1/3 of Earth per satellite; 3 GEOs give near-global coverage)
Path loss	Lowest (satellite is close)	Medium	Highest (satellite is very far)
Example use	Iridium, Starlink, remote sensing	GPS, GLONASS, O3b	TV broadcast, weather, fixed telephony

Feature	LEO (Low Earth Orbit)	MEO (Medium Earth Orbit)	GEO (Geostationary Orbit)
Tracking needed at earth station?	Yes (fast-moving)	Yes	No (fixed look angle once pointed)

Key trade-off to remember for essay questions: GEO trades delay and path loss for simplicity (no tracking, few satellites needed, always-on coverage), while LEO trades a more complex, expensive constellation and inter-satellite handover for low delay and low path loss.

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3. Orbital Mechanics — Kepler's Laws (Worked Numericals)

Two of Kepler's laws show up directly as calculation questions.

Kepler's Second Law — "Equal Areas in Equal Time"

A line joining a planet (or satellite) to the body it orbits sweeps out **equal areas in equal time intervals**. This means the sweep rate (area per unit time) is constant for a given orbit:

$$\text{Sweep rate} = \frac{\text{Area}_1}{\text{Time}_1} = \frac{\text{Area}_2}{\text{Time}_2}$$

Worked Example (matches your paper): *"A planet sweeps an area of 7.60×10¹⁸ km² in 32.5 hours. How long does it take to sweep an area of 3.20×10¹⁸ km²?"*

$$\text{Sweep rate} = \frac{7.60 \times 10^{18}}{32.5} = 2.338 \times 10^{17}\ \text{km}^2/\text{hr}$$

$$t = \frac{3.20 \times 10^{18}}{2.338 \times 10^{17}} \approx 13.69\ \text{hours}$$

Method to remember: find the constant rate from the given pair, then divide the new area by that rate.

Newton's Law of Gravitation

$$F = \frac{G\, m_1 m_2}{r^2}$$

where $G = 6.674 \times 10^{-11}$ N·m^2/kg^2, r = distance between centers (in metres).

Worked Example (matches your paper): *"Two spherical objects of equal mass experience a gravitational force of 25 N. Their centers are 36 cm apart. Find each mass."*

Since masses are equal, $m_1 = m_2 = m$, and $r = 0.36$ m:

$$F = \frac{Gm^2}{r^2} \implies m^2 = \frac{Fr^2}{G} = \frac{25 \times (0.36)^2}{6.674 \times 10^{-11}} = \frac{3.24}{6.674 \times 10^{-11}} = 4.855 \times 10^{11}$$

$$m = \sqrt{4.855 \times 10^{11}} \approx 6.97 \times 10^5 \text{ kg } (\approx 697{,}000 \text{ kg each})$$

Method to remember: rearrange for m^2 first when both masses are equal, then square-root at the end.

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4. Look Angle Calculation (Elevation & Azimuth)

The **look angle** tells an earth station installer exactly which direction to point a dish: how high above the horizon (**elevation**) and in which compass direction (**azimuth**). This is *the* signature GEO calculation question.

Given: - Earth station latitude L, longitude l_E - Satellite longitude l_S (GEO satellites sit at latitude 0°)

Step 1 — Relative longitude:

$$B = |l_E - l_S|$$

Step 2 — Elevation angle:

$$E = \arctan\left[\frac{\cos B \cos L - 0.1513}{\sqrt{1 - \cos^2 B \cos^2 L}}\right]$$

(0.1513 is the constant $R_e/(R_e + h)$ for a GEO satellite — Earth radius 6378 km over GEO radius 42,164 km.)

Step 3 — Azimuth angle — first find the intermediate angle:

$$A' = \arctan\left[\frac{\tan B}{\sin L}\right]$$

Then apply this rule (earth station in **Northern hemisphere**): - If satellite is **East** of earth station: Azimuth $= 180° + A'$ - If satellite is **West** of earth station: Azimuth $= 180° - A'$

(For a **Southern hemisphere** earth station: East → Azimuth $= A'$; West → Azimuth $= 360° - A'$.)

Worked Example (matches your paper): *Earth station: 52°N, 0° longitude. Satellite: 0° latitude, 66°E longitude.*

$$B = |0 - 66| = 66^\circ$$

$$\cos B = \cos 66^\circ = 0.4067, \quad \cos L = \cos 52^\circ = 0.6157$$

$$\cos B \cos L = 0.2504$$

$$E = \arctan\left[\frac{0.2504 - 0.1513}{\sqrt{1 - 0.2504^2}}\right] = \arctan\left[\frac{0.0991}{0.9681}\right] = \arctan(0.1024) \approx 5.85^\circ$$

$$A' = \arctan\left[\frac{\tan 66^\circ}{\sin 52^\circ}\right] = \arctan\left[\frac{2.246}{0.788}\right] = \arctan(2.850) \approx 70.7^\circ$$

Satellite (66°E) is **East** of the earth station (0°), and station is in the Northern hemisphere, so:

$$\text{Azimuth} = 180^\circ + 70.7^\circ = 250.7^\circ$$

Answer: Elevation ≈ 5.85°, Azimuth ≈ 250.7°

Method to remember: relative longitude → elevation formula → intermediate azimuth angle A′ → apply the hemisphere/direction rule. Always sanity-check: elevation angles well below ~5° are often considered impractical (signal blocked by horizon/buildings) — this is a good essay point about GEO limitations at high latitudes.

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5. Satellite System Architecture

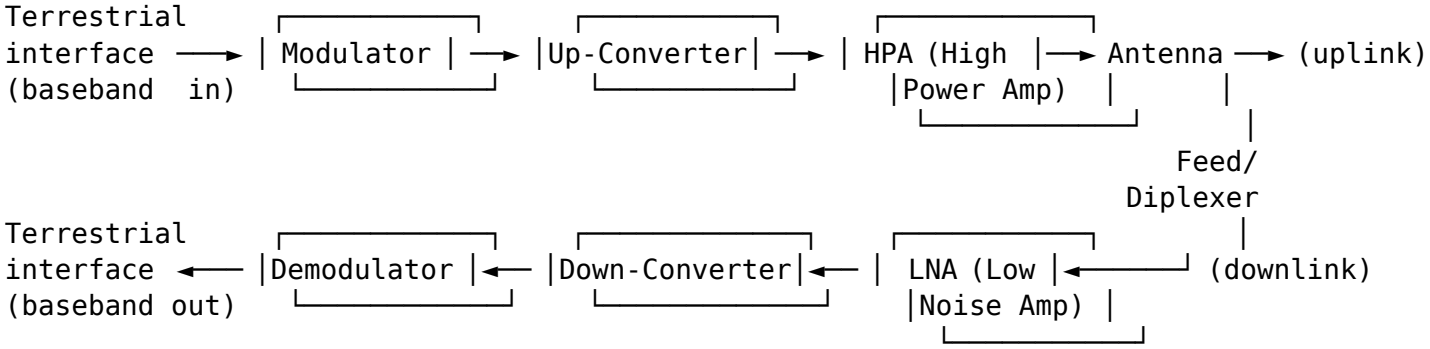
5.1 Space Segment — Parts of a Satellite

Bus (platform) — supports the payload; includes: - Structure (frame) - Power subsystem (solar arrays + batteries) - Attitude and orbit control (keeps antennas pointed correctly) - Propulsion (station-keeping thrusters) - Thermal control - **TT&C subsystem** (see §5.4)

Payload — the part that actually does the communication job; includes the **transponders** (see §5.3) and antennas.

5.2 Ground Segment — Earth Station Architecture (Block Diagram, in words)

A typical earth station has these functional blocks:



- **Antenna subsystem** – parabolic reflector + feed; shared for transmit and receive via a diplexer.
- **Transmit chain** – baseband signal → modulator → up-converter (to uplink RF frequency) → High Power Amplifier (HPA/TWTA) → antenna.
- **Receive chain** – antenna → Low Noise Amplifier (LNA, critical for a good G/T) → down-converter → demodulator → baseband output.
- **Tracking subsystem** – for non-GEO or large dishes, keeps antenna pointed at a moving satellite.
- **Power supply** – mains + backup.
- **Terrestrial interface** – connects the station to telephone exchange, internet backbone, TV studio, etc.

5.3 Satellite Transponder — Functions

A transponder is the "repeater channel" inside the payload. Its jobs, in order:

1. **Receive** the weak uplink signal via the receive antenna.
2. **Filter** it (input filter selects only the channel of interest, rejects noise/adjacent channels).
3. **Low-noise amplify** the very weak signal (LNA).
4. **Frequency-translate (down-convert)** from uplink frequency to downlink frequency — this avoids the transmitted downlink signal interfering with the received uplink signal.
5. **High-power amplify** (using a TWTA – Travelling Wave Tube Amplifier, or solid-state power amplifier) to a level strong enough to be usable after the huge downlink path loss.
6. **Retransmit** on the downlink antenna.

A single satellite carries many transponders (each handling a slice of bandwidth, e.g., 36 MHz), which is why "how many channels can be multiplexed onto a transponder" is a recurring theme.

5.4 TT&C Subsystem (Telemetry, Tracking & Command)

TT&C is how ground controllers keep the satellite alive and correctly positioned — separate from the communications payload traffic.

- **Telemetry (T)**: the satellite continuously sends **housekeeping data** to the ground — battery voltage, temperature, fuel level, attitude sensor data, subsystem status. This lets ground controllers monitor satellite health.
- **Tracking (T)**: ground stations measure the satellite's **range, range-rate, and angular position** (via ranging signals/Doppler) to determine and predict its exact orbit.
- **Command (C)**: ground controllers send **uplinked control instructions** to the satellite — e.g., fire thrusters for station-keeping, switch on/off transponders, reconfigure the payload, execute orbital maneuvers.

Simple block diagram (in words):

```
Ground TT&C Station                                  Satellite
 Command Encoder ── uplink ──► Command Receiver ──► Command Decoder ──► executes on subsyste
 Telemetry Decoder ◄── downlink ── Telemetry Transmitter ◄── Telemetry Encoder ◄── sensors/
 Tracking/Ranging ◄──────── ranging tones/beacon ────────► Transponder (beacon)
```

TT&C typically uses a dedicated narrowband frequency, separate from the main communication payload, so control of the satellite is never lost even if traffic transponders fail.

5.5 Satellite Power System — Solar Cells

- Solar arrays use the **photovoltaic effect**: photons striking a semiconductor (usually silicon or gallium arsenide) knock electrons loose, generating a DC current.
- Arrays are sized to supply peak load plus margin, and to recharge batteries.
- **Batteries** (typically nickel-hydrogen or lithium-ion) supply power during **eclipse periods**, when the Earth blocks sunlight from reaching the satellite (this happens for GEO satellites around the equinoxes).
- A **power conditioning/distribution unit** regulates and routes power to the payload and bus subsystems.

5.6 Active vs Passive Satellites

	Active Satellite	Passive Satellite
Does it amplify the signal?	**Yes** — receives, amplifies, retransmits (needs onboard power & electronics)	**No** — merely reflects the incoming signal without amplification
Onboard power needed?	Yes (solar/battery to run transponders)	Minimal/none
Example	Virtually all modern comsats (Telstar onward)	Echo 1 (a metallized balloon reflector, 1960)

	Active Satellite	Passive Satellite
Signal quality	Strong, usable downlink	Extremely weak (huge path loss both ways, no gain) — impractical for modern use

5.7 Functions of the Overall Satellite Communication System (Component Summary)

Component	Function
Earth Station	Transmits uplink signal, receives downlink signal, interfaces to terrestrial networks
Satellite Transponder	Receives, filters, amplifies, frequency-translates, and retransmits signals (§5.3)
Uplink Subsystem	The chain (modulator → up-converter → HPA → antenna) that gets the signal from Earth to the satellite
Downlink Subsystem	The chain (antenna → LNA → down-converter → demodulator) that gets the signal from satellite to Earth
Network/Satellite Control Center (NCC/SCC)	Monitors and controls the satellite (via TT&C), manages network resources, handles demand assignment of channels

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6. Frequency Bands Used in Satellite Communication

Band	Frequency Range (approx.)	Typical Use	Advantages	Disadvantages
L-Band	1–2 GHz	Mobile satellite (GPS, Inmarsat)	Low rain attenuation, penetrates foliage/weather well	Limited bandwidth, congested, needs larger antennas for gain

Band	Frequency Range (approx.)	Typical Use	Advantages	Disadvantages
S-Band	2–4 GHz	TT&C, some mobile/broadcast (satellite radio)	Low rain fade, good for mobile links	Limited bandwidth, shared with terrestrial services
C-Band	4–8 GHz	Fixed satellite service (FSS), TV distribution, VSAT	Very low rain fade — reliable in tropical/heavy-rain regions	Requires large dish (~2–3 m+), heavy terrestrial interference (shared with microwave links)
X-Band	8–12 GHz	Military & government communication	Good balance of bandwidth vs rain fade, resistant to jamming (military use)	Restricted to government/military allocation, not for commercial use
Ku-Band	12–18 GHz	DTH TV broadcast, VSAT	Smaller, cheaper dishes; more available bandwidth than C-band	More susceptible to **rain fade/rain attenuation**
Ka-Band	26.5–40 GHz	High-throughput broadband satellites	Very large bandwidth, very small dishes possible	Severe rain fade, requires advanced fade-mitigation (ACM), more expensive equipment

General rule for essay questions: as frequency increases (L→S→C→X→Ku→Ka), **available bandwidth increases** and **required antenna size decreases**, but **susceptibility to rain attenuation increases** and equipment cost/complexity rises. This trade-off is the standard “why not just use the highest frequency band” answer.

Impact on global communication (developing countries angle): higher bands (Ku/Ka) let developing countries deploy cheap, small VSAT terminals without expensive large dishes, accelerating rural connectivity — but heavy rainfall regions (much of the tropics) must budget extra rain margin or fall back to C-band for reliability, which needs bigger, costlier antennas.

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7. Intra-Orbital vs Inter-Orbital Links, and Routing

- **Intra-orbital link (intra-plane ISL)**: a link between two satellites that are in the **same orbital plane/shell** (e.g., two neighbouring LEO satellites in the same constellation ring). These links are relatively stable because the satellites maintain a roughly constant relative geometry.
- **Inter-orbital link (inter-plane ISL)**: a link between satellites in **different orbital planes or altitudes** (e.g., a LEO satellite linking to a satellite in an adjacent orbital plane, or a LEO-to-GEO relay). These links are more complex — the relative geometry between the two satellites constantly changes, requiring continuous antenna re-pointing/handover.

Routing in satellite networks: instead of every signal having to go back down to a ground gateway and back up again, modern constellations (especially large LEO systems) route traffic **satellite-to-satellite** across inter-satellite links until it reaches a satellite above the destination, which then downlinks locally.

Advantages of routing (in-orbit) in satellite systems: - Reduces dependence on a dense network of ground gateways (cheaper infrastructure, useful over oceans/remote regions with no ground stations). - Can reduce end-to-end latency versus multiple ground bounces (space is a near-vacuum, so light travels faster in vacuum than through fiber in some long-haul cases). - Improves resilience — traffic can be re-routed around a failed satellite or congested gateway. - Enables truly global coverage even where no local ground infrastructure exists.

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8. Multiple Access Techniques (FDMA, TDMA, CDMA)

Multiple access lets **many earth stations share one transponder** simultaneously. This is thoroughly covered in your Lecture 8 slides — summarized here in exam form.

8.1 FDMA — Frequency Division Multiple Access

- The transponder's bandwidth is divided into separate **frequency slots**; each earth station transmits continuously on its own assigned frequency.
- Users need only acquire the bandwidth/power they require; no time coordination needed — just stay on your assigned frequency.
- **Drawback**: multiple carriers sharing one amplifier (TWTA) cause **intermodulation distortion (IMD)**, so the amplifier must be run below saturation (back-off), wasting some power efficiency.
- Common in SCPC (single channel per carrier), VSAT data networks, older analogue video links.

8.2 TDMA — Time Division Multiple Access

- All earth stations share **one frequency** but transmit in **non-overlapping time slots**, in a repeating frame.
- Fully digital: data is buffered, then transmitted in a high-speed burst during the station's assigned slot.
- Requires precise **frame synchronization** (a reference/sync burst) so bursts from different stations don't overlap at the satellite.
- **Advantage**: the transponder's HPA only ever amplifies one carrier at a time → can run at full saturation → no IMD problem, more efficient power use than FDMA.
- A well-designed TDMA network can reach ~90% throughput efficiency (loss is only guard time + synchronization overhead).
- Time delay is roughly proportional to frame length / number of stations sharing the channel, since each station must wait its turn.

8.3 CDMA — Code Division Multiple Access

- All users transmit **on the same frequency, at the same time**, but each is assigned a **unique spreading code** (pseudorandom sequence).
- Two forms: **Direct Sequence (DSSS)** — most common in satellite systems — and **Frequency Hopping (FHSS)**.
- The receiver uses correlation with the matching code to extract the wanted signal; other users' signals appear as noise-like interference.
- **Advantages**: no need for tight time/frequency coordination; inherent security (hard to intercept without knowing the code); resistant to jamming (originally developed for military satellite use); graceful degradation.
- **Disadvantage**: capacity is not fixed — every additional active user raises the effective noise floor for everyone else (self-interference), so capacity is interference-limited, and careful **power control** is essential (a strong CDMA carrier can degrade every other user).

8.4 Demand Assignment (DAMA)

- **Demand-Assigned Multiple Access (DAMA)**: instead of permanently pre-allocating a frequency/timeslot/code to a station whether it has traffic or not (**pre-assigned/fixed access, "PAMA"**), the network assigns capacity to a station **only when it actually needs to transmit**, then releases it for others when done.
- Works alongside any of FDMA/TDMA/CDMA, but is most often associated with FDMA (assigning a channel on demand for the duration of a call, similar to a phone exchange).
- **Benefit**: much more efficient use of scarce transponder bandwidth when traffic is bursty (e.g., voice calls, not all stations transmitting all the time) — you only pay/use resources when actually communicating.

8.5 Satellite Packet Communication — ALOHA

- Designed for **bursty, low-duty-cycle data traffic** where reserving a dedicated slot (as in classic TDMA) would waste capacity.
- Stations transmit **whenever they have data**, without waiting for a coordinated turn — this can cause **collisions** if two stations transmit at overlapping times.
- A receiving hub sends an **acknowledgement**; if none is received within a time-out, the sender retransmits after a **random back-off delay** (to avoid repeated collisions).
- **Pure ALOHA** maximum theoretical throughput is only about 18%; **Slotted ALOHA** (transmissions aligned to time slots) roughly doubles this to ~37%.
- **Advantage**: minimal delay for light/bursty traffic (no waiting for a scheduled slot), simple to implement, no central scheduling needed for light loads.
- **Disadvantage**: throughput collapses under heavy load because collisions multiply.

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9. Traffic Multiplexing Techniques & Network Configurations

9.1 Multiplexing recap (from your general telecom notes — FDM/TDM/WDM/CDM)

- **FDM** – signals share a medium by occupying **different frequency slots** simultaneously (needs guard bands between channels).
- **TDM** – signals share a medium by taking turns in **different time slots** on the same frequency.
- **WDM** – conceptually FDM applied to light — different wavelengths (colours) of light carry separate channels on one optical fiber.
- **CDM** – signals share by using **different orthogonal/pseudorandom codes** on the same time and frequency (the basis of CDMA).

In satellite systems, these are the underlying multiplexing techniques that are then combined with a multiple-access scheme (FDMA uses FDM, TDMA uses TDM, CDMA uses CDM) to let many independent earth stations use a shared transponder.

9.2 Network Configurations

Configuration	Description	Typical Example
Point-to-point	One transmitting station communicates with exactly one receiving station	A dedicated trunk link between two telephone exchanges via satellite
Point-to-multipoint	One station transmits to many receiving stations simultaneously	A broadcast hub sending TV/data to many VSAT terminals (a “star” network)

Configuration	Description	Typical Example
Multipoint-to-point	Many stations transmit toward a single central receiving station	Many remote VSATs sending data back to a central hub
Multipoint-to-multipoint	Many stations can communicate with many other stations directly	A full **mesh** VSAT network where any site can talk directly to any other site

Note the pairing: point-to-multipoint and multipoint-to-point together describe a classic **star/hub-and-spoke VSAT network** (hub broadcasts out point-to-multipoint; remotes report back multipoint-to-point). A **mesh** network is multipoint-to-multipoint.

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10. Satellite Link Design & Link Budget (the core numeric skill)

A **link budget** is simply a careful accounting of every gain and loss a signal experiences from the transmitter to the receiver, so you can predict whether the received signal is strong enough (relative to noise) for the system to work.

10.1 Key Quantities & Formulas

EIRP (Effective Isotropic Radiated Power) — how strong the transmitted signal effectively is, accounting for antenna gain:

$$\text{EIRP(dBW)} = P_t\text{(dBW)} - L_{tx\ waveguide}\text{(dB)} + G_t\text{(dB)}$$

Free-Space Path Loss — power lost simply from spreading out over distance d at frequency f:

$$L_p\text{(dB)} = 32.45 + 20\log_{10}(f_{MHz}) + 20\log_{10}(d_{km})$$

(equivalently $L_p = 20\log_{10}(4\pi d/\lambda)$)

Received Carrier Power:

$$C\text{(dBW)} = \text{EIRP(dBW)} - L_p\text{(dB)} - L_{atm}\text{(dB)} + G_r\text{(dB)} - L_{rx\ waveguide}\text{(dB)}$$

Flux Density at a distance d (power spread over the surface of a sphere of radius d):

$$\Phi = \frac{\text{EIRP (linear, W)}}{4\pi d^2} \quad \left[\text{W/m}^2\right]$$

Received Power from Flux Density and effective aperture area A_e**:**

$$P_r = \Phi \times A_e$$

Relationship between gain and effective aperture area:

$$G = \frac{4\pi A_e}{\lambda^2} \quad \Longleftrightarrow \quad A_e = \frac{G\lambda^2}{4\pi}$$

Earth Station Figure of Merit, G/T (dB/K):

$$\frac{G}{T}(\text{dB/K}) = \left[G_r(\text{dB}) - L_{rx\ waveguide}(\text{dB})\right] - 10\log_{10}(T_{sys})$$

(if system noise temperature is already given in **dBK**, just subtract it directly)

Noise Power:

$$N(\text{dBW}) = k(\text{dBW/Hz/K}) + T_{sys}(\text{dBK}) + B(\text{dBHz})$$

where Boltzmann's constant $k = -228.6$ dBW/Hz/K.

Carrier-to-Noise Ratio:

$$\frac{C}{N}(\text{dB}) = C(\text{dBW}) - N(\text{dBW})$$

> **Tip:** Working entirely in **dB** turns every multiplication/division in the linear formulas into simple addition/subtraction — always convert to dB first, then just add/subtract down the link budget table.

10.2 Worked Example A — Flux Density & Received Power (matches your paper)

"A satellite at a distance of 39,000 km radiates a power of 20 W from an antenna with a gain of 22 dB. Effective aperture area at the receiving building is 10 m². Find (i) flux density, (ii) received power. If the satellite operates at 11 GHz and the Earth Station antenna has a gain of 52.3 dB, determine the received power."

Step 1 — EIRP:

$$P_t = 10\log_{10}(20) = 13.0 \text{ dBW}, \quad \text{EIRP} = 13.0 + 22 = 35.0 \text{ dBW } (\approx 3162 \text{ W linear})$$

Step 2 — Flux density ($d = 3.9 \times 10^7$ m):

$$4\pi d^2 = 4\pi(3.9 \times 10^7)^2 = 1.912 \times 10^{16} \text{ m}^2$$

$$\Phi = \frac{3162}{1.912 \times 10^{16}} = 1.65 \times 10^{-13} \text{ W/m}^2 \ \ (\text{or } -127.8 \text{ dBW/m}^2)$$

Step 3 — Received power (using 10 m² aperture):

$$P_r = \Phi \times A_e = 1.65 \times 10^{-13} \times 10 = 1.65 \times 10^{-12} \text{ W } \ (\approx -117.8 \text{ dBW})$$

Step 4 — Received power using Earth Station gain instead of aperture area (11 GHz, G_r = 52.3 dB):

Cross-check with Friis in dB form:

$$L_p = 32.45 + 20\log_{10}(11000) + 20\log_{10}(39000) = 32.45 + 80.83 + 91.82 = 205.1 \text{ dB}$$

$$P_r(\text{dBW}) = \text{EIRP} + G_r - L_p = 35.0 + 52.3 - 205.1 = -117.8 \text{ dBW}$$

Both methods agree: received power ≈ −117.8 dBW (≈1.66×10⁻¹² W). This cross-check (aperture-area method vs. gain/path-loss method) is exactly the kind of consistency check examiners like to see in your working.

10.3 Worked Example B — Full Link Budget Table (matches your paper)

Item	Link Parameter	Value	Units
1	Transmit power	10.0	dBW
2	Transmit waveguide loss	1.5	dB
3	Transmit antenna gain	27.0	dBi
4	Free space loss	196.0	dB
5	Atmospheric absorption	0.1	dB
6	Received antenna gain	40.2	dBi
7	Received waveguide loss	0.5	dB
8	System noise temperature	21.5	dBK
9	Boltzmann constant	−228.6	dBW/Hz/K
10	Bandwidth	74.0	dB Hz

(i) Satellite EIRP:

$$\text{EIRP} = 10.0 - 1.5 + 27.0 = 35.5 \text{ dBW}$$

(ii) Received carrier power:

$$C = 35.5 - 196.0 - 0.1 + 40.2 - 0.5 = -120.9 \text{ dBW}$$

(iii) Earth Station G/T:

$$G/T = (40.2 - 0.5) - 21.5 = 18.2 \text{ dB/K}$$

(iv) Noise power:

$$N = -228.6 + 21.5 + 74.0 = -133.1 \text{ dBW}$$

(v) Carrier-to-Noise ratio:

$$C/N = -120.9 - (-133.1) = 12.2 \text{ dB}$$

Method to remember: EIRP = Tx power − Tx waveguide loss + Tx antenna gain → Received carrier = EIRP − all path/atmospheric losses + Rx gain − Rx waveguide loss → G/T = (Rx gain − Rx waveguide loss) − noise temp(dBK) → Noise power = k + T(dBK) + B(dBHz) → C/N = Carrier − Noise. This exact five-line sequence answers almost any link-budget-table question in this course.

10.4 Worked Example C — Free-Space Path Loss & Received Power (matches your paper)

"Transmit power = 25 W, transmit antenna gain = 42 dB, frequency = 14 GHz, distance = 38,000 km, received antenna gain = 38 dB. Calculate (b) free-space path loss, (c) received power."

$$P_t(\text{dBW}) = 10\log_{10}(25) = 14.0 \text{ dBW}, \quad \text{EIRP} = 14.0 + 42 = 56.0 \text{ dBW}$$

(b) Free-space path loss ($f = 14{,}000$ MHz, $d = 38{,}000$ km):

$$L_p = 32.45 + 20\log_{10}(14000) + 20\log_{10}(38000) = 32.45 + 82.92 + 91.60 \approx 207.0 \text{ dB}$$

(c) Received power:

$$P_r(\text{dBW}) = \text{EIRP} - L_p + G_r = 56.0 - 207.0 + 38.0 = -113.0 \text{ dBW} \approx 5.0\times10^{-12} \text{ W (5 pW)}$$

10.5 The "Satellite Link Design Principle" (essay-style answer)

When explaining the *principle* behind designing a satellite link (rather than solving a numeric example), structure your answer around:

1. **Define the requirement** — required C/N (or E_b/N_0) for the chosen modulation/coding to achieve an acceptable bit error rate.
2. **Build the uplink budget** — Earth station EIRP, uplink path loss, satellite receive G/T → uplink C/N.
3. **Build the downlink budget** — satellite EIRP, downlink path loss, earth station receive G/T → downlink C/N.
4. **Combine uplink and downlink C/N** (they combine reciprocally, since noise adds at each stage) to get overall link C/N.
5. **Add a link margin** to protect against rain fade, pointing errors, and equipment ageing — especially critical at Ku/Ka band.
6. **Compare available C/N against required C/N** — if available ≥ required (with margin), the link design is acceptable; otherwise increase power, antenna gain, or use a more robust (but lower-rate) modulation/coding scheme.

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11. Quick-Reference Formula Sheet

Quantity	Formula
Kepler's 2nd law	$\text{Area}_1/\text{Time}_1 = \text{Area}_2/\text{Time}_2$
Gravitation	$F = Gm_1m_2/r^2$
Relative longitude	$B = \lvert l_E - l_S \rvert$
Elevation angle	$E = \arctan\left[\dfrac{\cos B\cos L - 0.1513}{\sqrt{1-\cos^2 B\cos^2 L}}\right]$

Quantity	Formula
Azimuth intermediate	$A' = \arctan\left[\dfrac{\tan B}{\sin L}\right]$
EIRP	$\text{EIRP(dBW)} = P_t - L_{tx} + G_t$
Free-space loss	$L_p\text{(dB)} = 32.45 + 20\log f_{MHz} + 20\log d_{km}$
Flux density	$\Phi = \dfrac{\text{EIRP (W)}}{4\pi d^2}$
Gain ↔ aperture	$G = \dfrac{4\pi A_e}{\lambda^2}$
Received power	$P_r = \text{EIRP} - L_p - L_{atm} + G_r - L_{rx}$
G/T	$(G_r - L_{rx}) - T_{sys}\text{(dBK)}$
Noise power	$N = k + T_{sys}\text{(dBK)} + B\text{(dBHz)}$, $k = -228.6$ dBW/Hz/K
C/N	$C\text{(dBW)} - N\text{(dBW)}$

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12. Rapid-Fire Answer Guide to Every Past Question

#	Question (paraphrased)	Where to find the answer here
1	Frequency bands (L/S/C/X/Ku) — discuss + pros/cons	§6
2	Planet sweeping area / Kepler's 2nd law calculation	§3
3	Gravitational force / mass calculation	§3
4	Earth station architecture + block diagram	§5.2
5	Active vs passive satellites	§5.6
6	Satellite flux density / received power calculation	§10.2
7	Intra-orbital vs inter-orbital link	§7
8	Link budget table (EIRP, carrier power, G/T, noise, C/N)	§10.3
9	Demand Assignment strategies	§8.4
10	Payload & transponder functions	§5.3

#	Question (paraphrased)	Where to find the answer here
11	Solar cells / power system	§5.5
12	TT&C subsystem + block diagram	§5.4
13	Satellite link design principle	§10.5
14	FDMA / CDMA / TDMA (with diagrams)	§8.1–8.3
15	Satellite packet communication (ALOHA)	§8.5
16	Satellite communication overview (how it works)	§1
17	LEO / MEO / GEO comparison	§2
18	Look angle of a GEO satellite	§4
19	Advantage of routing in satellite systems	§7
20	Routing concept in satellite communication	§7
21	Role of a transponder	§5.3
22	History of the satellite industry	§1.1
23	Traffic multiplexing techniques diagram	§9.1
24	Point-to-point / point-to-multipoint / multipoint-to-point / multipoint-to-multipoint	§9.2
25	Free-space path loss & received power calculation	§10.4
26	Functions of earth station, transponder, uplink/downlink subsystem, network control center	§5.7
27	Impact of frequency bands on global/developing-country communication	§6
28	Comparison/contrast of GEO, MEO, LEO orbits	§2

Good luck — you've got this. Focus your remaining time on §4 (look angle) and §10

(link budgets): once the formula sequence is automatic, those marks are the easiest to secure under time pressure.